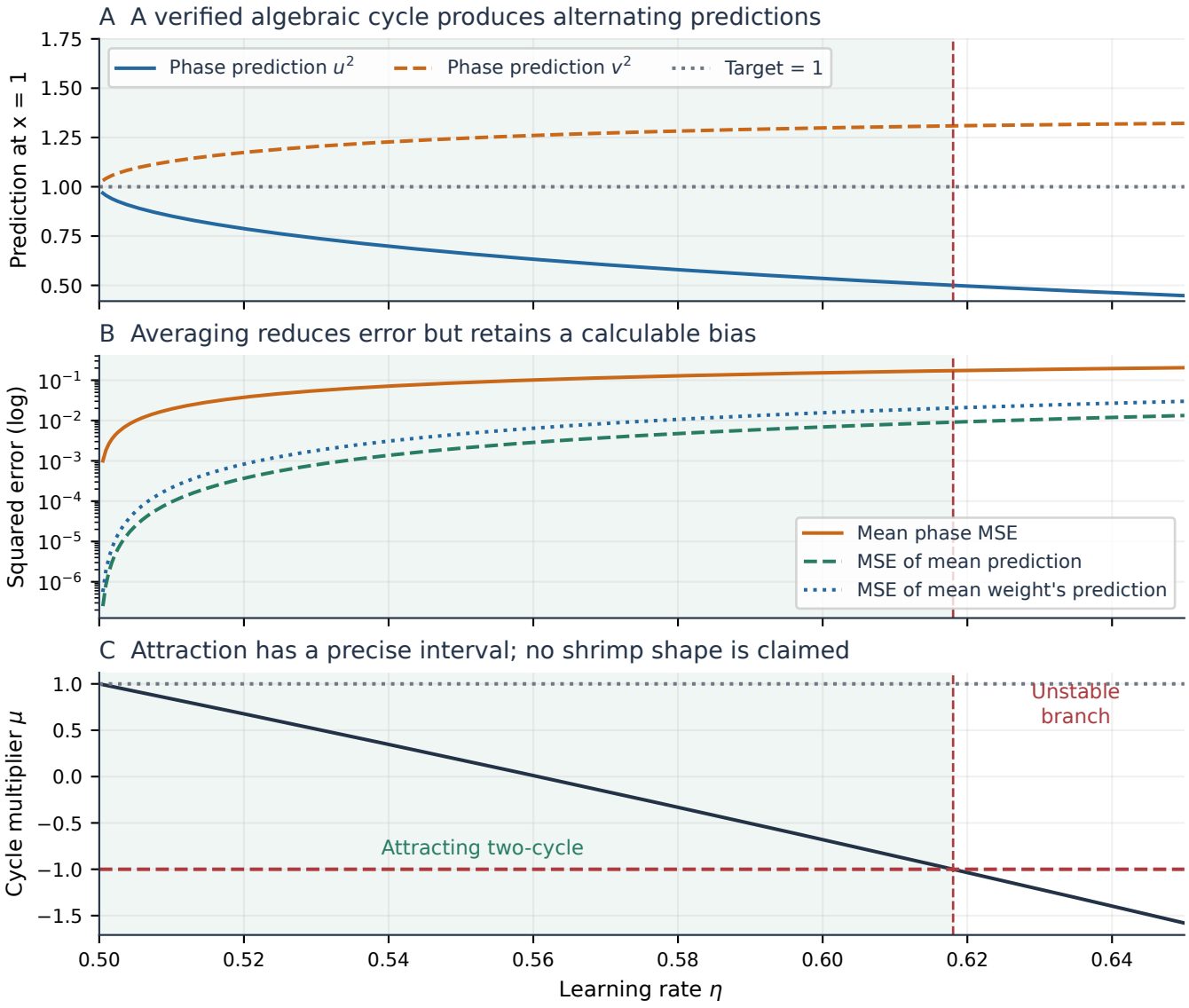
Exact toy neural training: $\hat{y} = \theta^2 x$, one sample $(x, y) = (1, 1)$
A A verified algebraic cycle produces alternating predictions
Prediction at x = 1
Phase prediction u^2
Phase prediction v^2
Target = 1
1.75
1.50
1.25
1.00
0.75
0.50
B Averaging reduces error but retains a calculable bias
Squared error (log)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
Mean phase MSE
MSE of mean prediction
MSE of mean weight's prediction
C Attraction has a precise interval; no shrimp shape is claimed
Cycle multiplier μ
1.0
0.5
0.0
−0.5
−1.0
−1.5
Unstable branch
Attracting two-cycle
0.50
0.52
0.54
0.56
0.58
0.60
0.62
0.64
Learning rate η